

Question ID: 1b1deebe

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

$ax + by = 72$

$6x + 2by = 56$

In the given system of equations, a and b are constants. The graphs of these equations in the xy -plane intersect at the point $(4, y)$. What is the value of a ?

Answer

- A. 3
- B. 4
- C. 6
- D. 14

Correct Answer: D

Rationale

Choice D is correct. It’s given that the graphs of the given system of equations intersect at the point $(4, y)$. Therefore, $(4, y)$ is the solution to the given system. Multiplying the first equation in the given system by -2 yields $-2ax - 2by = -144$. Adding this equation to the second equation in the system yields $(-2a + 6)x + (-2b + 2b)y = (-144 + 56)$, or $(-2a + 6)x = -88$. Since $(4, y)$ is the solution to the system, the value of a can be found by substituting 4 for x in this equation, which yields $(-2a + 6)(4) = -88$. Dividing both sides of this equation by 4 yields $-2a + 6 = -22$. Subtracting 6 from both sides of this equation yields $-2a = -28$. Dividing both sides of this equation by -2 yields $a = 14$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.